

Exponential ExpressionsQuestion Bank - Additional 3.03 **ANSWER KEY***Show your work. Circle your final answer.*

1. Two students evaluated $\left(\frac{3^5}{3^2}\right)^2$. Rylan first wrote $\frac{3^5}{3^2} = 3^3$, then $(3^3)^2 = 3^6 = 729$. Liam first wrote $\frac{3^{10}}{3^4}$, then $3^6 = 729$.

a. Name the law Rylan used first, and the law he used second.

quotient of powers, then power of a power

b. Name the law Liam used first, and the law he used second.

power of a quotient (with power of a power on top and bottom), then quotient of powers

c. List the laws needed to evaluate $(4^3 \cdot 4^0)^2$, then evaluate it.

zero exponent (or product of powers), then power of a power; $4^6 = 4,096$

For questions 2 through 6, use the laws of exponents to evaluate. Show each law you use.

2. $\left(\frac{2 \cdot 1^4}{2 \cdot 1^2}\right)^1$

$2 \cdot 1^2 = 4,41$

3. $\left[\left(\frac{3}{4}\right)^3 \cdot \left(\frac{3}{4}\right)^0\right]^2$

$\left(\frac{3}{4}\right)^6 = \frac{729}{4096}$

Math Nation's key prints 729/4,046 for this item; $4^6 = 4,096$.

4. $\frac{(4^2)^7}{4^{14}}$

$\frac{4^{14}}{4^{14}} = 4^0 = 1$

5. $\frac{3.3}{4 \cdot 1^0} \cdot 3.3^2$

$3.3 \cdot 3.3^2 = 3.3^3 = 35.937$

6. $\frac{(-6)^2 \cdot (-6)^1}{(-6)^0}$

$(-6)^3 = -216$

Equivalent or not.

7. Select ALL of the expressions that are equivalent to $\frac{0.3^2}{3^2} \cdot 0.1^3$.

- ☒ **A.** $\left(\frac{0.3}{3}\right)^2 \cdot 0.1^3$
☐ B. 0.1^6
☐ C. $0.3^0 \cdot 0.1^3$
☒ **D.** 0.1^5
☐ E. $0.3^2 \cdot 0.1^3$
☒ **F.** $0.1^2 \cdot 0.1^3$

A, D and F

$$0.1^2 \cdot 0.1^3 = 0.1^5 = 0.00001.$$

8. Which expression is equivalent to $\left(\frac{7^6}{7^4}\right)^3$?

- ☒ **A.** 7^6
☐ B. 7^{30}
☐ C. 7^5
☐ D. 7^2

A

Working backwards.

9. Find the value of n that makes each statement true.

a. $\left(\frac{3^n}{3^2}\right)^2 = 3^8$

$n = 6$

b. $\frac{2^n \cdot 2^0}{2^4} = 8$

$n = 7$

10. Kim says that $\frac{(-18)^3}{6^3}$ can be rewritten as $\left(\frac{-18}{6}\right)^3$ before evaluating. Is she right? Name the law, and give the value.

Yes — power of a quotient, read from right to left; $(-3)^3 = -27$.